Female Infertility

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Microbiome Signatures, Microbiome Medicine

Abstract. Microbiome imbalances contribute to female infertility and targeted interventions, such as probiotics, diet, and lifestyle changes, are offering new solutions to enhance fertility and reproductive health.

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Overview

Female infertility is a complex and multifactorial condition that affects approximately 10-15% of women of reproductive age globally. It is defined as the inability to conceive after one year of unprotected intercourse, though this rate increases with age, particularly after 35 years.[1] In the United States, around 1 in 8 couples experience infertility, with the female factor contributing to nearly 40-50% of cases. Common causes of female infertility include ovulatory disorders, structural abnormalities of the reproductive organs, and conditions like Polycystic Ovary Syndrome (PCOS), Endometriosis, and Pelvic Inflammatory Disease (PID). Bacterial Vaginosis (BV), which affects 29% of women of reproductive age, is also a significant contributor, as its association with chronic pelvic inflammatory disease and endometritis can lead to infertility.[2] In addition to infections, hormonal imbalances, autoimmune diseases, and environmental factors also contribute to female infertility. Recent studies have also highlighted the crucial role of the microbiome in both the vaginal and gut environments, with disruptions in the microbiome contributing to inflammation, immune system alteration, and infertility.[3][4] With advancing age, the chances of successful conception drop, and fertility treatments like In-Vitro Fertilization (IVF) are often required. The success rates for IVF vary but are generally higher for women under 35, with success rates declining to approximately 10-15% by age 40.[5] Collectively, infertility is a major public health concern, and there is a need for early diagnosis, intervention, and treatment to improve reproductive outcomes, minimize the psychological and emotional burden on affected individuals, and reduce healthcare costs associated with prolonged infertility treatments.

Associated Conditions

Polycystic Ovary Syndrome (PCOS) is a common condition linked to infertility and characterized by hormonal imbalances, including hyperandrogenism and anovulation, which disrupt ovulation and hinder fertility. Women with PCOS are also at risk for insulin resistance and metabolic dysfunction, further complicating their reproductive health.[6] Another significant contributor to infertility is Pelvic Inflammatory Disease (PID), often caused by infections such as *Chlamydia trachomatis* and *Neisseria gonorrhoeae*. PID leads to inflammation in the upper reproductive tract, resulting in complications like

tubo-ovarian abscesses and fallopian tube damage, which can directly impair fertility.[7] Studies show that chronic PID and repeated infections increase the risk of tubal factor infertility.[8]Endometriosis is another major cause of infertility, affecting nearly 50% of women with the condition.[9] It disrupts ovarian reserve, reduces embryo quality, and alters endometrial receptivity, significantly lowering the chances of conception. The pathophysiology of endometriosis-related infertility is multifactorial, involving chronic inflammation, immune system alterations, and physical disruptions of reproductive organs.[10]Bacterial Vaginosis (BV), characterized by an imbalance in the vaginal microbiome with a reduction in *Lactobacillus* species and an increase in anaerobic bacteria like *Gardnerella vaginalis*, is closely associated with infertility. BV can lead to pelvic inflammatory disease and chronic endometritis, further impacting fertility by causing inflammation in the uterus and fallopian tubes.[11]

Causes

The pathogenesis of infertility remains complex and is influenced by a variety of factors. One of the primary theories is infection-mediated damage, where ascending pathogens such as *Chlamydia trachomatis* and *Gardnerella vaginalis* induce PID.[12] This inflammation can lead to scarring, tubal occlusion, and damage to reproductive organs, resulting in infertility. While infection is a major contributor to infertility in some cases, many instances remain unexplained, particularly when no overt infection is present. Another theory involves immune modulation by the microbiome. Dysbiosis, which refers to an imbalance in microbial communities, can alter the mucosal immune system.[13] This shift in immune tone may lead to conditions like chronic endometritis, which impairs endometrial receptivity and the chances of successful implantation. A third theory focuses on endocrine disruption, where changes in the microbiome of the gut and reproductive tract can affect hormone metabolism.[15] Some research points to metabolomic effects, where microbial metabolites are absorbed into the bloodstream and may directly impact ovarian function, folliculogenesis, and embryo development.[16] These theories, while promising, still have limitations, as the precise mechanisms linking the microbiome to infertility are not yet fully understood, and much of the evidence remains correlative rather than causal.

Diagnosis

Diagnosis of infertility increasingly incorporates microbiome and metabolomic signatures as promising non-invasive methods. One of the most relevant techniques is microbiome profiling, which uses 16S rRNA sequencing to analyze the vaginal microbiome.[17] This method has shown that a Lactobacillus-dominated microbiota is often associated with better reproductive outcomes, while a dysbiotic microbiome, typically characterized by pathogens like *Gardnerella vaginalis* and *Atopobium vaginae*, is linked to infertility.[18] The gut microbiome also plays a role in infertility diagnosis. Studies using Mendelian randomization have identified specific gut microbiota, such as *Faecalibacterium*, which are associated with increased risk of reproductive tract inflammation and infertility.[19][20] In addition to microbiome analyses, recent advancements in miRNA biomarkers have shown potential. For example, elevated levels of miR-21-5p and miR-155-5p in vaginal and rectal samples have been linked to infertility caused by microbiome-induced inflammation.[21] These promising diagnostic approaches offer the potential for early detection of infertility risk and for tailoring interventions aimed at restoring microbial balance to improve fertility outcomes.

Primer

A fundamental understanding of female infertility requires a solid grasp of how the microbiome intersects with reproductive health. In the female reproductive system, the microbiome predominantly resides in the vaginal and uterine environments. It is typically dominated by beneficial *Lactobacillus* species, which contribute to an acidic environment that prevents pathogen overgrowth and supports immune function.[22] However, when this microbial balance is disrupted it has been linked to infertility through mechanisms such as inflammation and immune system alterations.[23] For instance, dysbiosis in the vaginal microbiome, characterized by an overgrowth of pathogens like *Gardnerella vaginalis*, has been associated with pelvic inflammatory disease and endometriosis, both of which can compromise fertility.[24] Recent studies have revealed the presence of microbiomes in the upper reproductive tract, such as the uterus and Fallopian tubes, which were previously considered sterile. [25]These microbial communities are thought to influence reproductive health by modulating the immune response and hormone metabolism, further complicating the role of the microbiome in fertility. The gut microbiome plays a role in modulating systemic inflammation and metabolic processes that can impact the reproductive system. Understanding these microbiome–reproductive system interactions is essential for developing targeted therapies, such as probiotics or other microbiome-targeted interventions, to restore microbial balance and improve fertility outcomes.

Metallomic Signatures

Heavy metals such as lead (Pb), cadmium (Cd), and arsenic (As) are significant environmental pollutants that have been implicated in female infertility. These metals can accumulate in the human body through various sources, including contaminated food, water, air, and occupational exposures.[26] The toxic effects of these metals on the reproductive system are profound, as they disrupt several biological processes crucial for fertility. For example, lead has been associated with menstrual cycle disturbances, hormonal imbalances, and reduced ovarian function.[27] Cadmium, a known endocrine disruptor, impacts ovarian follicle development, induces oxidative stress, and disrupts hormonal regulation, while arsenic exposure can lead to oxidative damage and affect oocyte maturation.[28][29] The cumulative impact of these metals is often seen in impaired fertility, with a higher risk of conception delays, miscarriage, and reduced pregnancy success rates. Epidemiological studies, including those based on NHANES data, have demonstrated that even low levels of exposure to these metals are associated with increased infertility risks.

▶ **What is the metallomic of Female Infertility?**

Metabolomic Signatures

Metabolomic profiling has become a crucial tool in understanding female infertility, as it offers insights into the biochemical environment of the follicular fluid (FF) and its role in oocyte development and fertilization. Studies have shown that women with infertility often exhibit altered metabolic signatures in FF, including imbalances in antioxidants, energy metabolites, and markers of oxidative stress.[47] Key compounds such as lactate, ammonia, and creatinine are found in higher concentrations in the FF of infertile women, while glucose and amino acids, which are vital for oocyte quality, are often reduced.[48] Oxidative stress markers like malondialdehyde (MDA) and reduced glutathione (GSH) are altered, reflecting the negative impact of oxidative damage on oocyte health and IVF success rates.[49] These metabolic changes are particularly evident in conditions such as polycystic ovary syndrome (PCOS), endometriosis, and age-related reduced ovarian reserve, where distinct metabolic profiles could help

predict IVF outcomes.[50] Environmental factors, such as exposure to endocrine-disrupting chemicals like bisphenol A (BPA) and di(2-ethylhexyl) phthalate (DEHP), contribute to infertility by disrupting hormonal regulation and increasing oxidative stress, further complicating fertility issues.[51] The cumulative effects of these pollutants, which interfere with metabolic processes critical for reproductive health, have been shown to significantly increase infertility risk.

Interventions

Microbiome-targeted interventions (MBTIs) are emerging as an essential element in the treatment of infertility, especially in addressing conditions like Bacterial Vaginosis (BV), Polycystic Ovary Syndrome (PCOS), Endometriosis, and Pelvic Inflammatory Disease (PID). These conditions have been increasingly associated with microbial dysbiosis, a disruption in the balance of the vaginal and uterine microbiomes, which can lead to infertility. MBTIs aim to restore healthy microbial communities, promote optimal immune responses, and prevent microbial overgrowth, which can be pivotal in enhancing fertility outcomes. In light of these developments, MBTIs are gaining significant attention in clinical research as potential preventative and therapeutic tools for improving reproductive health.

Intervention	Mechanism	MBTI Status
Medical-Grade Honey (MGH)	MGH creates an osmotic environment that inhibits bacterial growth, reduces inflammation, and promotes the growth of Lactobacillus species, the key beneficial bacteria that maintain vaginal health.[52] Honey also exhibits anti-inflammatory and antioxidant properties, which support a healthy vaginal microbiome. These combined effects make it effective for preventing recurrent BV and improving fertility outcomes by restoring a balanced microbiome.[53]	Experimental
Probiotics	Probiotics, particularly Lactobacillus strains, promote vaginal microbiome restoration by enhancing lactic acid production, which lowers vaginal pH and suppresses harmful pathogens.[54] They also produce bacteriocins, natural antibiotics that fight harmful microorganisms.[55] By restoring a Lactobacillus-dominant microbiome, probiotics enhance fertility and support healthy reproductive tract environments.[56]	Promising Candidate
Metformin	Metformin indirectly supports fertility in women with Polycystic Ovary Syndrome (PCOS) by improving insulin sensitivity. Insulin resistance is common in PCOS and negatively affects ovarian function. By improving insulin regulation, metformin helps restore gut microbiota balance, which influences hormonal regulation.[57]	Promising Candidate
Vaginal Microbiota Transplantation (VMT)	VMT introduces healthy, donor-derived microbiota into the vaginal environment. This transplant restores a balanced microbiome, reducing inflammation and promoting healthy reproductive function. By replenishing Lactobacillus, it prevents the overgrowth of harmful bacteria, supporting fertility.[58][59]	Experimental
Mediterranean Diet	The Mediterranean Diet has systemic effects on the gut microbiome, which influences female reproductive health. Rich in polyunsaturated fatty acids (PUFAs), antioxidants, and fiber, this diet promotes healthy gut microbiota that regulates immune responses, reduces systemic inflammation, and improves hormonal regulation.[60] These changes in the gut microbiome help support ovarian function, improve endometrial receptivity, and optimize the chances of successful conception.[61]	Promising Candidate
Dietary Fiber (Guar Gum)	Dietary fiber, specifically Guar Gum, functions as a prebiotic to promote the growth of beneficial gut bacteria.The effects of guar gum positively influences endometrial receptivity, ovarian reserve, and menstrual regularity, all of which are critical factors for fertility. Guar Gum's ability to reduce gut dysbiosis makes it a potential therapy for improving fertility outcomes, especially in women undergoing ART.[62]	Under Investigation

STOPs

The STOPs initiative encourages the critical reassessment of common everyday practices that may quietly undermine reproductive health and contribute to female infertility. Many widely accepted habits, from dietary choices to herbal remedies, are rarely questioned, yet they can significantly impact hormonal balance, ovulation, and the delicate interplay of factors necessary for conception. One example is the regular consumption of hibiscus tea. Often promoted for its antioxidant and blood pressure-lowering effects, hibiscus also contains natural phytoestrogens, plant compounds that can mimic or disrupt normal estrogen function.[63] Emerging research suggests that frequent intake may influence menstrual cycle length, progesterone levels, and ovulatory patterns, potentially lowering the chances of successful conception in some women.

STOP	Microbiome Impact	InfertilityRisk
Hibiscus rosa-sinensis	Alter gut and vaginal microbiome composition through phytoestrogen activity, though evidence is still emerging.[64][65][66]	Disrupts hormonal balance (estrogen/progesterone), may interfere with ovulation and menstrual cycle regularity, potentially reducing conception rates.[67]

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